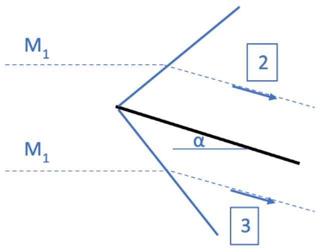
AERO50001 Aerodynamics 2 Problem Set #7

Extra practice: Linear supersonic theory

7.1 Flat plate

★★☆ 8-15 mins

Use Ackeret's theory to estimate the pressure coefficient on each side of a flat plate in a flow at $M_1 = 2.5$ at an incidence angle of $3°$.



(a) At the top of the plate, C_{p2}.

(b) At the bottom of the plate, C_{p3}

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(a) $C_{p2} = \frac{2(-\theta)}{\sqrt{M_1^2 - 1}} = -0.0457$

(b) $C_{p3} = -\frac{2\theta^{(-\theta)}}{\sqrt{M_1^2 - 1}} = \overset{+}{-}0.0457$